

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006, as amended by Commission Regulation (EU) 2020/878

Version 6.11
Revision Date 10.07.2026
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GENERIC EU MSDS - NO COUNTRY SPECIFIC DATA - NO OEL DATA

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifiers

Product name : Diethyl ether

Product Number : 346136
Brand : SIGALD
Index-No. : 603-022-00-4
REACH No. : 01-2119535785-29-XXXX
CAS-No. : 60-29-7

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laboratory chemicals, Manufacture of substances

1.3 Details of the supplier of the safety data sheet

Company : Merck Life Science LLC
Valovaya 35
115054 MOSCOW
RUSSIAN FEDERATION

Telephone : +7 7 495 621-5828
Fax : +7 7 495 621-6037

1.4 Emergency telephone number

Emergency Phone # : +8(800) 100-63-46 (CHEMTREC)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Flammable liquids, Category 1	H224: Extremely flammable liquid and vapour.
Acute toxicity, Category 4	H302: Harmful if swallowed.
Specific target organ toxicity - single exposure, Category 3, Central nervous system	H336: May cause drowsiness or dizziness.



Long-term (chronic) aquatic hazard,
Category 3

H412: Harmful to aquatic life with long
lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms

:



Signal word

:

Danger

Hazard statements

:

H224 Extremely flammable liquid and vapour.
H302 Harmful if swallowed.
H336 May cause drowsiness or dizziness.
H412 Harmful to aquatic life with long lasting effects.

Supplemental Hazard
Statements

:

EUH019 May form explosive peroxides.
EUH066 Repeated exposure may cause skin dryness
or cracking.

Precautionary
statements

:

Prevention:

P210 Keep away from heat, hot surfaces, sparks,
open flames and other ignition sources. No
smoking.
P233 Keep container tightly closed.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/
eye protection/ face protection.

Response:

P301 + P312 IF SWALLOWED: Call a POISON CENTER/
doctor if you feel unwell.

Storage:

P403 + P233 Store in a well-ventilated place. Keep
container tightly closed.

Reduced Labelling (<= 125 ml)

Hazard pictograms

:



Signal word

Danger

Hazard Statements
H224

Extremely flammable liquid and vapour.



H412	Harmful to aquatic life with long lasting effects.
Precautionary Statements	
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233	Keep container tightly closed.
P280	Wear protective gloves/ protective clothing/ eye protection/ face protection.
P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
Supplemental Hazard information (EU)	
EUH019	May form explosive peroxides.
EUH066	Repeated exposure may cause skin dryness or cracking.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.1 Substances

Substance name	: Ether
Index-No.	: 603-022-00-4
EC-No.	: 200-467-2

Components

Chemical name	CAS-No. EC-No.	Concentration (% w/w)	M-Factor, SCL, ATE
Diethyl ether	60-29-7 200-467-2	>= 90 - <= 100	specific concentration limit STOT SE 3; H336 >= 20 % Acute toxicity estimate Acute oral toxicity: 1.211 mg/kg
butyl hydroxytoluene	128-37-0	>= 1 - < 2,5	



SECTION 4: First aid measures

4.1 Description of first aid measures

General advice	: Show this safety data sheet to the doctor in attendance.
If inhaled	: After inhalation: fresh air. Call in physician.
In case of skin contact	: In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.
In case of eye contact	: After eye contact: rinse out with plenty of water. Remove contact lenses.
If swallowed	: After swallowing: immediately make victim drink water (two glasses at most). Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

4.3 Indication of any immediate medical attention and special treatment needed

No data available

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	: Carbon dioxide (CO ₂) Foam Dry powder
Unsuitable extinguishing media	: For this substance/mixture no limitations of extinguishing agents are given.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire fighting	: Combustible. Pay attention to flashback. Vapours are heavier than air and may spread along floors. Development of hazardous combustion gases or vapours possible in the event of fire. Forms explosive mixtures with air at ambient temperatures.
Hazardous combustion products	: Carbon oxides



5.3 Advice for firefighters

- Special protective equipment for fire-fighters : In the event of fire, wear self-contained breathing apparatus.
- Further information : Remove container from danger zone and cool with water.
Prevent fire extinguishing water from contaminating surface water or the ground water system.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

- Personal precautions : Advice for non-emergency personnel:
Do not breathe vapours, aerosols.
Avoid substance contact.
Ensure adequate ventilation.
Keep away from heat and sources of ignition.
Evacuate the danger area, observe emergency procedures, consult an expert.

For personal protection see section 8.

6.2 Environmental precautions

- Environmental precautions : Do not let product enter drains.
Risk of explosion.

6.3 Methods and material for containment and cleaning up

- Methods for cleaning up : Cover drains. Collect, bind, and pump off spills.
Observe possible material restrictions (see sections 7 and 10).
Take up with liquid-absorbent material (e.g. Chemizorb®). Dispose of properly. Clean up affected area.

6.4 Reference to other sections

For disposal considerations see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

- Advice on safe handling : Work under hood. Do not inhale substance/mixture.
Avoid generation of vapours/aerosols.
- Advice on protection against fire and explosion : Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharge.
- Hygiene measures : Change contaminated clothing. Preventive skin protection recommended. Wash hands after working with substance.



Temperature class : T4

For precautions see section 2.2.

7.2 Conditions for safe storage, including any incompatibilities

Further information on storage conditions : Keep container tightly closed in a dry and well-ventilated place. Keep away from heat and sources of ignition.

Storage class (TRGS 510) : 3, Flammable liquids

Further information on storage stability : Light sensitive.
Heat sensitive.
Air sensitive.
Test for peroxide formation periodically and before distillation.

Recommended storage temperature : Recommended storage temperature see product label.

Packaging material : Suitable material: Mild Steel Drum

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
Ether	Fresh water	2 mg/l
	Marine water	0,2 mg/l
	Fresh water sediment	9,14 mg/kg
	Marine sediment	0,914 mg/kg
	Soil	0,66 mg/kg

8.2 Exposure controls

Individual protection measures, such as personal protective equipment (PPE)

Eye/face protection : Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).
Safety glasses

Hand protection

Material : Viton®
Break through time : 30 min
Glove thickness : 0,7 mm
Protective index : Splash contact
Manufacturer : Vitoject® (KCL 890 / Aldrich Z677698, Size M)



- Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-36124 Eichenzell, Internet: www.kcl.de).
- Skin and body protection : Flame retardant antistatic protective clothing.
- Respiratory protection : required when vapours/aerosols are generated.
- Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter type: : Filter type AX
type:
The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer. These measures have to be properly documented.

Environmental exposure controls

Advice : Do not let product enter drains.
Risk of explosion.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

- Physical state : liquid (20 °C, 1.013 hPa)
- Colour : colourless
- Odour : sweet, ether-like
- Melting point/ range : -116 °C
Method: lit.
- Boiling point/boiling range : 34,6 °C
Method: lit.
- Flammability : No data available
- Upper explosion limit /
Upper flammability limit : Upper explosion limit
36 %(V)
- Lower explosion limit / : Lower explosion limit



Lower flammability limit	1,7 %(V)
Flash point	: -40 °C
Autoignition temperature	: 180 °C Auto-flammability
Decomposition temperature	: No data available
pH	: No data available
Viscosity	
Viscosity, dynamic	: 0,195 mPa.s (40 °C)
Viscosity, kinematic	: No data available
Flow time	: No data available
Solubility(ies)	
Water solubility	: 65 g/l (20 °C) pH: 7 completely soluble
Partition coefficient: n-octanol/water	: log Pow: 1,1 Bioaccumulation is not expected.
Vapour pressure	: 189 hPa (0 °C) 389 hPa (10 °C) 563 hPa (20 °C) 863 hPa (30 °C) 1.228 hPa (40 °C) 2.311 hPa (60 °C)
Relative density	: No data available
Density	: 0,706 g/mL (25 °C)



Method: lit.

Relative vapour density : 2,56
(Air = 1.0)

9.2 Other information

Explosives : Not classified as explosive.

Oxidizing properties : none

Burning rate : No data available

Self-ignition : 175 °C
ca. 1.013,25 hPa

Evaporation rate : No data available

Molecular weight : 74,12 g/mol

SECTION 10: Stability and reactivity

10.1 Reactivity

Formation of peroxides possible.
Vapours may form explosive mixture with air.

10.2 Chemical stability

The product is chemically stable under standard ambient conditions (room temperature) .
Contains the following stabiliser(s):
butyl hydroxytoluene (BHT)(≤11 %)

10.3 Possibility of hazardous reactions

Hazardous reactions : Risk of ignition or formation of inflammable gases or vapours with:
chromyl chloride
Peroxides
Risk of explosion with:
azides
halogens
halogen-halogen compounds
nonmetallic oxyhalides
Strong oxidizing agents
chromium(VI) oxide
halogen oxides
peroxi compounds
perchloric acid
perchlorates



Nitric acid
nitrating acid
Oxygen
Ozone
turpentine oils and/or turpentine substitutes
nitrates
metallic chlorides
salts of oxyhalogenic acids
nitrogen oxides
nonmetallic oxides
chromosulfuric acid
chlorates
hydrogen peroxide
permanganic acid
sulfuric acid
with
Nitric acid
sulfur
Risk of explosion during distillation.
Exothermic reaction with:

acid halides

10.4 Conditions to avoid

Conditions to avoid : Light.
Heat
Air

Warming.

Moisture.

10.5 Incompatible materials

No data available

10.6 Hazardous decomposition products

Peroxides
In the event of fire: see section 5

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Acute toxicity estimate Oral - 1.224 mg/kg
(Calculation method)

LD50 Oral - Rat - 1.211 mg/kg (Diethyl ether)

Remarks: (RTECS)

Symptoms: Risk of aspiration upon vomiting., Aspiration may cause pulmonary oedema and pneumonitis.



Acute toxicity estimate Oral - 1.211 mg/kg (Diethyl ether)
(ATE value derived from LD50/LC50 value)
LC50 Inhalation - Mouse - 4 h - 97,5 mg/l - vapour
(Diethyl ether)
Remarks: (RTECS)
Symptoms: mucosal irritations
LD50 Dermal - Rabbit - male - > 20.000 mg/kg (Diethyl ether)
(OECD Test Guideline 402)
Remarks: (ECHA)

Skin corrosion/irritation

Skin - Rabbit (Diethyl ether)
Result: No skin irritation - 4 h
(OECD Test Guideline 404)
Remarks: Dermatitis

Serious eye damage/eye irritation

Eyes - Rabbit (Diethyl ether)
Result: No eye irritation
(OECD Test Guideline 405)

Respiratory or skin sensitization

Local lymph node assay (LLNA) - Mouse (Diethyl ether)
Result: negative
(OECD Test Guideline 429)

Germ cell mutagenicity

Test Type: Mouse
Test system: Embryo
Remarks: DNA inhibition
Test Type: Micronucleus test
(Diethyl ether)
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 487
Result: negative
Test Type: In vitro mammalian cell gene mutation test
(Diethyl ether)
Test system: Mouse lymphoma test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
Test Type: Ames test
(Diethyl ether)
Test system: Escherichia coli/Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
(Diethyl ether)
Test Type: Micronucleus test
Species: Mouse
Cell type: Red blood cells (erythrocytes)
Application Route: Intraperitoneal
Method: OECD Test Guideline 474
Result: negative



Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

Oral - May cause drowsiness or dizziness. - Central nervous system (Diethyl ether)

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information**Endocrine disrupting properties****Product:**

Assessment

The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Repeated dose toxicity - Rat - male and female - Oral - 13 Weeks - No observed adverse effect level - 500 mg/kg - Lowest observed adverse effect level - 2.000 mg/kg

Remarks: (ECHA)

(Diethyl ether)

RTECS: KI5775000

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

Inhalation may provoke the following symptoms: (Diethyl ether)

Cough, chest pain, Difficulty in breathing, Dizziness, Drowsiness, Contact with eyes can cause:, Redness, Provokes tears., Blurred vision, Prolonged or repeated exposure to skin causes defatting and dermatitis. (Diethyl ether)

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated. (Diethyl ether)

Narcotic!

(Diethyl ether)

After absorption:

(Diethyl ether)

Salivation

ataxia (impaired locomotor coordination)

inebriation

Unconsciousness

Coma

(Diethyl ether)

Other dangerous properties can not be excluded.

(Diethyl ether)

Handle in accordance with good industrial hygiene and safety practice.

(Diethyl ether)

Liver - Ingestion may provoke the following symptoms:, Irregularities - Based on Human Evidence



SECTION 12: Ecological information

12.1 Toxicity

Components:

Diethyl ether:

Toxicity to fish	: LC50 (<i>Lepomis macrochirus</i> (Bluegill sunfish)): > 10.000 mg/l Exposure time: 96 h Remarks: (IUCLID)
Toxicity to daphnia and other aquatic invertebrates	: EC50 (<i>Daphnia magna</i> (Water flea)): 1.380 mg/l Exposure time: 48 h Remarks: (IUCLID)
Toxicity to algae/aquatic plants	: ErC50 (<i>Desmodesmus subspicatus</i> (green algae)): > 100 mg/l Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes NOEC (<i>Desmodesmus subspicatus</i> (green algae)): 100 mg/l End point: Growth inhibition Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes
Toxicity to microorganisms	: EC50 (activated sludge): 21.000 mg/l Exposure time: 3 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 209 GLP: yes NOEC (activated sludge): 42 mg/l Exposure time: 3 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 209 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC: > 100 mg/l End point: reproduction rate Exposure time: 21 d Species: <i>Daphnia magna</i> (Water flea)



Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 211
GLP: yes

butyl hydroxytoluene (BHT):

- Toxicity to fish : LC50 (Danio rerio (zebra fish)): > 0,57 mg/l
End point: mortality
Exposure time: 96 h
Test Type: semi-static test
Analytical monitoring: yes
Method: Directive 67/548/EEC, Annex V, C.1.
GLP: yes
- Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0,48 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes
- Toxicity to algae/aquatic plants : ErC50 (Desmodesmus subspicatus (green algae)): > 0,4 mg/l
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: Regulation (EC) No. 440/2008, Annex, C.3
GLP: yes
- Toxicity to microorganisms : EC50 (activated sludge): > 10.000 mg/l
Exposure time: 3 h
Test Type: static test
Method: OECD Test Guideline 209
GLP: yes
- Toxicity to fish (Chronic toxicity) : NOEC: 0,053 mg/l
Exposure time: 30 d
Species: Oryzias latipes
Method: OECD Test Guideline 210
GLP: yes
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EC50: 0,096 mg/l
End point: reproduction rate
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Method: OECD Test Guideline 211
GLP: yes

Ecotoxicology Assessment

Acute aquatic toxicity : no acute aquatic toxicity



Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

12.2 Persistence and degradability

Components:

Diethyl ether:

Biodegradability : Remarks: Not readily biodegradable.

12.3 Bioaccumulative potential

Components:

Diethyl ether:

Bioaccumulation : Remarks: No bioaccumulation is to be expected (log Pow <= 4).

Partition coefficient: n-octanol/water : log Pow: 1,1
Remarks: Bioaccumulation is not expected.

butyl hydroxytoluene (BHT):

Partition coefficient: n-octanol/water : log Pow: 5,1
GLP: yes
Remarks: Potential bioaccumulation (ECHA)

12.4 Mobility in soil

No data available

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

Diethyl ether:

Assessment : Substance does not meet the criteria for PBT or vPvB according to Regulation (EC) No 1907/2006, Annex XIII.

butyl hydroxytoluene (BHT):

Assessment : Substance does not meet the criteria for PBT or vPvB according to Regulation (EC) No 1907/2006, Annex XIII.

12.6 Endocrine disrupting properties

Product:



Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Components:

butyl hydroxytoluene (BHT):

Additional ecological information : Discharge into the environment must be avoided.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14: Transport information

14.1 UN number or ID number

ADR : UN 1155

IMDG : UN 1155

IATA : UN 1155

14.2 UN proper shipping name

ADR : DIETHYL ETHER

IMDG : DIETHYL ETHER

IATA : Diethyl ether

14.3 Transport hazard class(es)

	Class	Subsidiary risks
ADR	: 3	
IMDG	: 3	
IATA	: 3	

14.4 Packing group

ADR
Packing group : I



Classification Code : F1
Hazard Identification : 33
Number
Labels : 3
Tunnel restriction code : (D/E)

IMDG

Packing group : I
Labels : 3
EmS Code : F-E, S-D

IATA (Cargo)

Packing instruction (cargo : 361
aircraft)
Packing group : I
Labels : Class 3 - Flammable liquids

IATA_P (Passenger)

Packing instruction : 351
(passenger aircraft)
Packing group : I
Labels : Class 3 - Flammable liquids

14.5 Environmental hazards

ADR

Environmentally : no
hazardous

IMDG

Marine pollutant : no

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing : Conditions of restriction for the
on the market and use of certain dangerous following entries should be
substances, mixtures and articles (Annex XVII) considered:
Number on list 3

Number on list 40
This substance/mixture shall
not be used in aerosol
dispensers intended for supply
to the general public for
entertainment and decorative



REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59).	purposes. : Not applicable
Regulation (EU) No 2024/590 on substances that deplete the ozone layer	: Not applicable
Regulation (EU) 2019/1021 on persistent organic pollutants (recast)	: Not applicable
REACH - List of substances subject to authorisation (Annex XIV)	: Not applicable

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances. P5a FLAMMABLE LIQUIDS

15.2 Chemical safety assessment

A Chemical Safety Assessment has been carried out for this substance.

SECTION 16: Other information

Full text of H-Statements

H336	: May cause drowsiness or dizziness.
EUH019	: May form explosive peroxides.
EUH066	: Repeated exposure may cause skin dryness or cracking.
H336	: May cause drowsiness or dizziness.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Flam. Liq.	: Flammable liquids
STOT SE	: Specific target organ toxicity - single exposure
STOT SE	: Specific target organ toxicity - single exposure

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous



Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information : The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sigma-aldrich.com and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.
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REG_EU / EN



Annex: Exposure scenario

Identified uses:

Use: Industrial use

SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
SU 3, SU9, SU 10: Industrial uses: Uses of substances as such or in preparations at industrial sites, Manufacture of fine chemicals, Formulation [mixing] of preparations and/ or re-packaging (excluding alloys)
PC19: Intermediate
PC21: Laboratory chemicals
PROC1: Use in closed process, no likelihood of exposure
PROC2: Use in closed, continuous process with occasional controlled exposure
PROC3: Use in closed batch process (synthesis or formulation)
PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises
PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)
PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities
PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities
PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
PROC10: Roller application or brushing
PROC15: Use as laboratory reagent
ERC1, ERC2, ERC4, ERC6a, ERC6b: Manufacture of substances, Formulation of preparations, Industrial use of processing aids in processes and products, not becoming part of articles, Industrial use resulting in manufacture of another substance (use of intermediates), Industrial use of reactive processing aids

Use: Professional use

SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
PC21: Laboratory chemicals
PROC15: Use as laboratory reagent
ERC2, ERC6a, ERC6b: Formulation of preparations, Industrial use resulting in manufacture of another substance (use of intermediates), Industrial use of reactive processing aids

1. Short title of Exposure Scenario: Industrial use

Main User Groups	: SU 3
Sectors of end-use	: SU 3, SU9, SU 10
Chemical product category	: PC19, PC21



Process categories : **PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC15**
 Environmental Release Categories : **ERC1, ERC2, ERC4, ERC6a, ERC6b:**

2. Exposure scenario

2.1 Contributing scenario controlling environmental exposure for: ERC1, ERC4

Other given operational conditions affecting environmental exposure

Number of emission days per year : 300

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as special waste in compliance with local and national regulations.

2.1 Contributing scenario controlling environmental exposure for: ERC2

Other given operational conditions affecting environmental exposure

Number of emission days per year : 300

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as special waste in compliance with local and national regulations.

2.1 Contributing scenario controlling environmental exposure for: ERC6a

Daily amount per site (Msafe) : 8.828 kg

Environment factors not influenced by risk management

Flow rate : 18.000 m3/d

Dilution Factor (River) : 10

Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 20

Emission or Release Factor: Air : 5 %

Emission or Release Factor: : 2 %

Water

Emission or Release Factor: Soil : 0,1 %

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Flow rate of sewage treatment plant effluent : 2.000 m3/d

Effectiveness (of a measure) : 89 %



Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as hazardous waste in compliance with local and national regulations.

2.1 Contributing scenario controlling environmental exposure for: ERC6b

Daily amount per site (Msafe) : 3.534 kg

Environment factors not influenced by risk management

Flow rate : 18.000 m³/d

Dilution Factor (River) : 10

Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per : 20

year

Emission or Release Factor: Air : 0,1 %

Emission or Release Factor: : 5 %

Water

Emission or Release Factor: Soil : 0,025 %

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Flow rate of sewage treatment : 2.000 m³/d

plant effluent

Effectiveness (of a measure) : 89 %

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as hazardous waste in compliance with local and national regulations.

2.5 Contributing scenario controlling worker exposure for: PROC1, PROC2**Product characteristics**

Concentration of the Substance in : Covers the percentage of the substance in the product
Mixture/Article up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor without local exhaust ventilation (LEV)

Enclosed process

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC3, PROC4, PROC8b**Product characteristics**

Concentration of the Substance in : Covers the percentage of the substance in the product
Mixture/Article up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor without local exhaust ventilation (LEV)
Enclosed process

Technical conditions and measures

Provide a good standard of general or controlled ventilation (5 to 15 air changes per hour).

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8a

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor with local exhaust ventilation (LEV)

Technical conditions and measures

Provide a good standard of general or controlled ventilation (5 to 15 air changes per hour).

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC9

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor with local exhaust ventilation (LEV)

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC5, PROC10

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure



Outdoor / Indoor : Indoor with local exhaust ventilation (LEV)

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC15

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor without local exhaust ventilation (LEV)

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR*
ERC1	ECETOC TRA		All compartments			< 0,01
ERC4	ECETOC TRA		All compartments			< 0,01
ERC2	ECETOC TRA		All compartments			< 0,01
ERC6a	ECETOC TRA		Soil		8828kg/day	1
ERC6b	ECETOC TRA		Soil		3534kg/day	1

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR*
PROC1	ECETOC TRA	longterm, inhalative, systemic			0,10
PROC1	ECETOC TRA	longterm, dermal,			0,03



		systemic			
PROC1	ECETOC TRA	longterm, combined, systemic			0,13
PROC2	ECETOC TRA	longterm, inhalative, systemic			0,10
PROC2	ECETOC TRA	longterm, dermal, systemic			0,03
PROC2	ECETOC TRA	longterm, combined, systemic			0,13

*Risk characterisation ratio

PROC3	ECETOC TRA	longterm, inhalative, systemic			0,98
PROC3	ECETOC TRA	longterm, dermal, systemic			0,01
PROC3	ECETOC TRA	longterm, combined, systemic			0,99
PROC4	ECETOC TRA	longterm, inhalative, systemic			0,01
PROC4	ECETOC TRA	longterm, dermal, systemic			0,30
PROC4	ECETOC TRA	longterm, combined, systemic			0,31
PROC8b	ECETOC TRA	longterm, inhalative, systemic			0,44
PROC8b	ECETOC TRA	longterm, dermal, systemic			0,16
PROC8b	ECETOC TRA	longterm, combined, systemic			0,60

*Risk characterisation ratio

PROC8a	ECETOC TRA	longterm, inhalative, systemic			0,49
PROC8a	ECETOC TRA	longterm, combined, systemic			0,80
PROC8a	ECETOC TRA	longterm, dermal, systemic			0,31

*Risk characterisation ratio

PROC9	ECETOC TRA	longterm,			0,20
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		inhalative, systemic			
PROC9	ECETOC TRA	longterm, dermal, systemic			0,02
PROC9	ECETOC TRA	longterm, combined, systemic			0,22

*Risk characterisation ratio

PROC5	ECETOC TRA	longterm, inhalative, systemic			0,25
PROC5	ECETOC TRA	longterm, dermal, systemic			0,002
PROC5	ECETOC TRA	longterm, combined, systemic			0,25
PROC10	ECETOC TRA	longterm, inhalative, systemic			0,25
PROC10	ECETOC TRA	longterm, dermal, systemic			0,031
PROC10	ECETOC TRA	longterm, combined, systemic			0,28

*Risk characterisation ratio

PROC15	ECETOC TRA	longterm, inhalative, systemic			0,49
PROC15	ECETOC TRA	longterm, dermal, systemic			0,01
PROC15	ECETOC TRA	longterm, combined, systemic			0,50

*Risk characterisation ratio

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For scaling of worker exposure assessments performed with ECETOC TRA, please consult the Merck tool ScIDeEx® at www.merckmillipore.com/scideex.

Please refer to the following documents: ECHA Guidance on information requirements and chemical safety assessment Chapter R.12: Use descriptor system; ECHA Guidance for downstream users; ECHA Guidance on information requirements and chemical safety assessment Part D: Exposure Scenario Building, Part E: Risk Characterisation and Part G: Extending the SDS; VCI/Cefic REACH Practical Guides on Exposure Assessment and Communications in the Supply Chain; CEFIC Guidance Specific Environmental Release Categories (SPERCs).



1. Short title of Exposure Scenario: Professional use

Main User Groups : **SU 22**
Sectors of end-use : **SU 22**
Chemical product category : **PC21**
Process categories : **PROC15**
Environmental Release Categories : **ERC2, ERC6a, ERC6b:**

2. Exposure scenario

2.1 Contributing scenario controlling environmental exposure for: ERC2

Other given operational conditions affecting environmental exposure

Number of emission days per : 300
year

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as special waste in compliance with local
and national regulations.

2.1 Contributing scenario controlling environmental exposure for: ERC6a

Daily amount per site (Msafe) : 8.828 kg

Environment factors not influenced by risk management

Flow rate : 18.000 m³/d
Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per : 20
year
Emission or Release Factor: Air : 5 %
Emission or Release Factor: : 2 %
Water
Emission or Release Factor: Soil : 0,1 %



Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant
Flow rate of sewage treatment : 2.000 m3/d
plant effluent
Effectiveness (of a measure) : 89 %

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as hazardous waste in compliance with local and national regulations.

2.1 Contributing scenario controlling environmental exposure for: ERC6b

Daily amount per site (Msafe) : 3.534 kg

Environment factors not influenced by risk management

Flow rate : 18.000 m3/d
Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per : 20
year
Emission or Release Factor: Air : 0,1 %
Emission or Release Factor: : 5 %
Water
Emission or Release Factor: Soil : 0,025 %

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant
Flow rate of sewage treatment : 2.000 m3/d
plant effluent
Effectiveness (of a measure) : 89 %

Conditions and measures related to external treatment of waste for disposal

Disposal methods : Dispose of as hazardous waste in compliance with local and national regulations.

2.4 Contributing scenario controlling worker exposure for: PROC15**Product characteristics**

Concentration of the Substance in : Covers the percentage of the substance in the product
Mixture/Article up to 100 % (unless stated differently).
Physical Form (at time of use) : High volatile liquid

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor without local exhaust ventilation (LEV)

Organisational measures to prevent /limit releases, dispersion and exposure

Covers daily exposures up to 8 hours.

Additional good practice advice beyond the REACH Chemical Safety Assessment

Wear suitable gloves (tested to EN374) and eye protection.

3. Exposure estimation and reference to its source**Environment**

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Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR*
ERC2	ECETOC TRA		All compartments			< 0,01
ERC6a	ECETOC TRA		Soil		8828kg/day	1
ERC6b	ECETOC TRA		Soil		3534kg/day	1

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR*
PROC15	ECETOC TRA	longterm, inhalative, systemic			0,5
PROC15	ECETOC TRA	longterm, dermal, systemic			0,008
PROC15	ECETOC TRA	longterm, combined, systemic			0,51

*Risk characterisation ratio

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For scaling of worker exposure assessments performed with ECETOC TRA, please consult the Merck tool ScIDeEx® at www.merckmillipore.com/scideex. Please refer to the following documents: ECHA Guidance on information requirements and chemical safety assessment Chapter R.12: Use descriptor system; ECHA Guidance for downstream users; ECHA Guidance on information requirements and chemical safety assessment Part D: Exposure Scenario Building, Part E: Risk Characterisation and Part G: Extending the SDS; VCI/Cefic REACH Practical Guides on Exposure Assessment and Communications in the Supply Chain; CEFIC Guidance Specific Environmental Release Categories (SPERCs).

